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A

PBL-II Report

ON

"CampusEats – Online Canteen Food Ordering For MET Campus"

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY
IN THE PARTIAL FULFILLMENT
OF

MASTER OF COMPUTER APPLICATION

FOR A.Y.2025–26

SUBMITTED BY

SANJANA BONDE	**ROLL NO: 07**
YASHIKA CHAWLA	**ROLL NO: 12**
HARSHAD DHONGADE	**ROLL NO: 16**
RAHUL HOWALE	**ROLL NO: 20**

UNDER THE GUIDANCE OF
Prof.Mehraj Khan

DEPARTMENT OF MCA
MET'S INSTITUTE OF ENGINEERING,
BHUJBAL KNOWLEDGE CITY,
ADGAON,NASHIK-422003

SAVITRIBAI PHULE PUNE UNIVERSITY

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Institute of Engineering,Nashik
Department of MCA
2025-26

CERTIFICATE

This is to certify that,

Sanjana Bonde	**Roll No.: 07**
Yashika Chawla	**Roll No.: 12**
Harshad Dhongade	**Roll No.: 16**
Rahul Howale	**Roll No.: 20**

We have successfully completed this PBL-II Report entitled "CampusEats – Online Canteen Food Ordering For MET Campus" under guidance of Prof.Mehraj Khan in partial fulfillment of the requirements for the Master degree of Computer Application under the Savitribai Phule Pune University during the academic year 2025-26

Prof.Mehraj Khan
Project Guide

Prof.P.D.Jadhav
HOD

Place: Nashik
Date:

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Abstract

In the modern digital era, technology has become imperative in bringing efficiency and convenience to all aspects of life. On a college campus, canteen management faces several problems such as long queues, poor communication between students and canteen staff, and delays in order processing. A solution to these problems comes through with **CampusEats** – an **Online Canteen Food Ordering For MET Campus** – designed to bring ease and digitization to the entire process of food ordering for students and canteen administrators.

The **CampusEats** is user-friendly, fast, and interactive, allowing students and staff to browse through the menu, check the availability of items, view prices, and place orders conveniently from any device. It eliminates the need to wait in long queues by enabling online ordering and tracking, so users can pick up their food as soon as it is ready. It also helps canteen staff handle orders, update menus, track sales, and minimize manual record-keeping tasks.

The platform provides powerful menu management, order monitoring, and payment tracking on the admin side, while on the user side, it ensures a seamless experience with login, signup, order history, and live order status. The website focuses on providing a smooth, responsive, and secure experience using modern web technologies like **HTML, CSS, JavaScript, PHP, MySQL, and Flask (Python)** for backend integration.

Implementing **CampusEats** in the canteen would replace manual processes with a modern, automated, and time-saving system. It would not only enhance user satisfaction but also improve canteen workflow efficiency. Overall, this project represents a step toward building a **smart campus environment** and contributing to the digital transformation of campus operations, demonstrating how technology can make life easier through innovation and teamwork.

INDEX

Topic	Page No.
Chapter 1: Introduction	4-6
1.1 Overview	4
1.2 Motivation	4-5
1.3 Objective	5
1.4 Organization of Report	6
Chapter 2: Literature Survey	7-9
2.1 Introduction	7
2.2 Existing Systems	7-8
2.3 Comparative Analysis	8-9
2.4 Proposed System	9
Chapter 3: Problem Statement	10
3.1 Problem Statement	10
Chapter 4: Project Requirements	11-22
4.1 System Features	11
4.2 Hardware Requirements	12
4.3 Software Requirements	13
4.4 Other Requirements	13
4.5 Database Requirements	14
4.6 Analysis Model	14-22
Chapter 5: System Analysis	23-26
5.1 System Architecture	23-24
5.2 Modules	24-25
5.3 Algorithm or Techniques Used	25-26
Chapter 6: Project Plan	27-33
6.1 Stakeholders List	27-28
6.2 System Implementation Plan	28-33
Chapter 7: Conclusion	34

Bibliography	35
List of Tables	
1. Comparative Analysis	9
2. Hardware Requirements	13
3. Software Requirements	14
4. Stakeholders List	28
List of Figures	
1. DFD Level 0	16
2. DFD Level 1 – User Side	17
3. DFD Level 1 – Admin Side	18
4. ER Diagram	19
5. Use Case Diagram	19
6. Sequence Diagram	20
7. Activity Diagram – Admin Side	21
8. Activity Diagram – User Side	22
9. Class Diagram	23

Chapter 1
Introduction

1.1 Overview

In the contemporary era of digitalization, technology has taken its core place in everyday life, bringing efficiency, comfort, and accessibility towards almost all spheres. From education and healthcare to transportation and communication, its effect can be well accounted for. One of those spheres where technology can serve a lot of convenience is food service management, especially on college campuses.

A college canteen is one of those places that serve many students and staff daily. Whenever orders are taken manually, a lot of problems arise, including long queues, time delays, order mix-ups, and difficulties in managing payments during rush hours. These issues cause a waste of time and create a problem for both the users and canteen staff.

These problems led to the development of **CampusEats – Online Canteen Food Ordering For MET Campus**. The purpose of this project, therefore, is to automate the processes of ordering food in college canteens. This website allows users to navigate through an easy digital platform where they could check the availability of menu items and prices, place an order, make online payments, and get the status for the placed order. It reduces reliance on manual mechanisms and saves time by providing a seamless and efficient workflow that enhances user experience.

The website has been designed keeping in mind simplicity, responsiveness, and accessibility. Students and staff could thereby access the website via their devices, whether laptops or mobile phones, thus making it a flexible solution suitable for any campus environment. The back-end part of the website allows the canteen management to efficiently manage orders, update menus, manage stock availability, and monitor order history through a secure and interactive interface.

Overall, **CampusEats** will transform the traditional canteen into a smart, organized, and time-saving digital platform. Indeed, this is in line with the growing trend towards automation and introducing smart campus initiatives.

1.2 Motivation

The idea of **CampusEats** came from observing the hustle experienced by students and canteen staff during rush hours. From long queues to miscommunication about orders, there are delays in service that contribute

to inefficiency and low customer satisfaction. With traditional ordering and management done manually, not only has much time been wasted but also a margin for errors was left.

Since most sectors were using technology for improved efficiency, it was very much expected that a food ordering platform would be introduced for the sake of convenience and ease to reduce the manual effort of one and all. The motivation for this project is to develop a system that saves time, reduces errors, and thereby enhances the overall user as well as canteen operator experience.

This inspired us, as students of the **MCA Department**, to utilize knowledge in web development, database management, and backend programming to provide a real-life solution that can be implemented within our college itself. **CampusEats** is an attempt at applying technical skills in solving a practical problem and hence making a contribution toward the digital growth of our institution.

The project also aligns with the visions of modern education: innovation, teamwork, and digital transformation of campus operations. In such a context, **CampusEats** could mean a small yet meaningful step toward constructing a smart campus ecosystem.

1.3 Objective

This project aims to design and develop a web-based application to simplify and automate the food ordering process in the campus canteen. Therefore, the major objectives of the **CampusEats** project are the following:

- To build an appealing, responsive, and user-friendly website for ordering food online.
- To eliminate long queues and minimize manual work in the canteen through an automated ordering process.
- To provide real-time capabilities in order tracking and notification.
- To allow the administrator to manage menus, track orders, update the availability of items, and keep sales records effectively.
- To ensure secure storage of data and communication between user interface and backend using technologies like PHP, MySQL, and Flask.
- To develop a scalable solution adaptable for other institutional or organizational food services.

By meeting these objectives, the project will not only increase the convenience of users but also enhance operational efficiency for the canteen management team.

1.4 Organization of Report

This project report is systematically divided into multiple chapters to present the work in a structured and comprehensive manner. Each chapter relates to a particular aspect of the project, right from conceptualization to the implementation and conclusion of it. The organization of the report is as follows:

- **Chapter 1: Introduction** – Introduces the background, motivation, objectives, and structure of the project report.
- **Chapter 2: Literature Survey** – Gives an overview of existing systems, research papers, and technologies concerning online food ordering, identifying the gap that this project fills.
- **Chapter 3: System Analysis and Design** – Describes system requirements, architecture, data flow diagrams, database design, and UI structure.
- **Chapter 4: Implementation and Results** – Explains project development, tools, testing, and the results obtained.
- **Chapter 5: Conclusion and Future Scope** – Summarizes findings and outlines possible future improvements.

This introductory chapter provides the foundation for understanding the need, purpose, and overall structure of the **CampusEats – Online Canteen Food Ordering For MET Campus** project. The coming chapters elaborate on design, functionality, and implementation details that collectively help bring this project into successful reality.

Chapter 2
Literature Survey

2.1 Introduction

Literature survey is one of the crucial activities that help in knowing the existing developments, research work, and technological trends on the subject. It helps to identify the strengths and weaknesses of current systems and forms a base for designing a better and innovative solution.

Food ordering systems online have gained much popularity in today's digital world because of their convenience, efficiency, and time-saving nature. Many research works and platforms have been developed to automate the process of ordering food, handling payments, and managing deliveries. However, most of the works regarding the development of these systems are for large-scale commercial purposes and not optimized for smaller environments such as college canteens.

The web-based **CampusEats – Online Canteen Food Ordering For MET Campus**, therefore, tries to fill this gap by studying the available systems for developing a model that meets the specific requirements of campus food services with criteria such as simplicity, real-time interaction, and cost-effectiveness.

2.2 Existing Systems

Over the years, several online food ordering systems were introduced to simplify the process of ordering food. Among the most successful ones are **Zomato**, **Swiggy**, and **Uber Eats**. These systems enable users to browse menus, select items, make payments online, and track delivery in real time. They use advanced technologies like GPS tracking, cloud databases, and secure payment gateways to ensure a smooth and reliable user experience.

Though these systems have proved to be very successful in large cities and commercial areas, they have a lot of limitations when it comes to implementation in a college canteen. First, they are designed for multiple restaurants and large customer bases, making them complex and costly for small-scale institutions. Second, they require delivery networks and third-party services, which are unnecessary within a campus. Thirdly, they demand continuous server support and maintenance, increasing operational costs.

The existing mechanisms in some colleges allow students to view menus and place orders using simple digital canteen systems. However, most of these systems lack essential features such as:

- Real-time order tracking,
- Dynamic menu updates,

- User authentication and secure login,
- Payment integration,
- Smooth communication between users and canteen staff.

Due to these limitations, there is a growing need for a solution that specifically addresses the requirements of college campuses—simple, efficient, and cost-effective, yet powerful enough to handle all food service operations digitally.

2.3 Comparative Analysis

To understand the advantages of the proposed system, a comparison has been made between existing commercial systems, basic college systems, and the proposed **CampusEats** model.

Features	**Existing Systems (Zomato / Swiggy)**	**Basic College Systems**	**Proposed System (CampusEats)**
Target Users	Whole World	Students	Students & Staff
Real-time Order Tracking	✓	×	✓
Menu Management	✓	Limited	✓
User Authentication	✓	×	✓
Cost & Maintenance	High	Low	Low
Admin Control Panel	✓	Few	✓
Customization for Campus	×	Partial	✓
Payment Handling	✓	×	✓
User Interface	Advanced	Basic	Interactive & Simple

Table : 1

From the comparison, it is evident that while commercial food delivery systems are feature-rich, they are too complex and expensive for a campus environment. On the other hand, basic canteen systems lack automation, scalability, and real-time functionality.

Thus, a hybrid model is necessary — one that combines the simplicity and affordability of basic systems with the robustness and automation of advanced ones. This is where **CampusEats** stands out as the perfect solution.

2.4 Proposed System (CampusEats)

CampusEats is a dedicated web-based system designed for automating and simplifying the food ordering process within a college campus. It bridges the gap between traditional manual order-taking methods and complex commercial food delivery applications.

In this system:

- Students and staff can register, login, browse menus, place orders, and track their order status in real time.
- The canteen staff can manage menus, update availability, process orders, and monitor sales efficiently via an admin dashboard.
- The interface is responsive, user-friendly, and accessible across multiple devices.

CampusEats is developed using modern web technologies such as **HTML, CSS, JavaScript, PHP, Flask, and MySQL**. The system ensures fast processing, data security, and reliable database management.

It reduces waiting time, minimizes human errors, and improves communication between users and canteen administrators. By implementing **CampusEats**, college canteens can embrace digital transformation, improve service quality, and enhance overall user satisfaction — contributing significantly to the concept of a **smart campus**.

Chapter 3
Problem Statement

3.1 Problem Statement

Most canteen managements on college campuses still depend on manual operations for taking and delivering orders of food items. The traditional approach leads to long queues, confusion in ordering, and wastage of time, particularly during the peak hours. Students experience a lot of inconvenience due to delayed service, while the canteen staff are not able to handle multiple orders effectively.

Manual procedures make tracking of payments, menu updating, and record maintenance very difficult. The system, without automation, lacks accuracy, transparency, and proper communication between students and canteen staff. Furthermore, record-keeping and report generation are time-consuming and prone to errors.

This means any existing commercial food delivery platform like **Zomato**, **Swiggy**, or **Uber Eats** would not be feasible for such a small-scale college canteen as they are complex, expensive, and feature-rich systems aimed at large-scale commercial environments. Thus, an affordable and customized web-based solution that caters to the specific needs of a campus environment is necessary.

The major concerns of the existing system are as follows:

- Long queues and waiting times during peak hours.
- Frequent order errors and a lack of real-time updates.
- Management issues in handling several orders simultaneously.
- No centralized system for managing menus, orders, and payments.
- Poor communication between students and canteen staff.
- Manual record-keeping and absence of digital reports.

These problems thus necessitated the development of the **CampusEats – Online Canteen Food Ordering For MET Campus**, which will provide an automated, efficient, and user-friendly online ordering platform, followed by digital tracking and management.

Faster services, improved accuracy, and better coordination are the key features this solution will ensure for a better canteen experience on campus.

Chapter 4
Project Requirements

4.1 System Features

CampusEats is an online canteen food ordering website developed to digitize the traditional canteen ordering system within the college campus. The system is divided into two modules: **User Side** for Students/Teachers and **Admin Side** for Canteen Administrator. Both modules are interconnected and managed through a secure MySQL database.

4.1.1 System Feature 1 – User Side (Functional Requirements)

The user module facilitates food ordering by students and teachers via a web-based platform, eliminating queues and improving efficiency. Focus is on responsive design, user-friendliness, and personalized experience.

Functional Requirements:

- **User Registration & Login:** New users can sign up using their name, email, password, and contact number. Existing users can log in securely.
- **Menu Browsing:** Users can explore food categories such as Snacks, Meals, Beverages, and Desserts with item details, prices, and images.
- **Favorites Management:** Users can mark food items as favorites for quick future access and reordering.
- **Cart Management:** Users can add multiple items to their cart, update quantities, and view totals before placing an order.
- **My Orders:** Displays all current and past orders with details like order ID, food items, total amount, and real-time status (Placed, Preparing, Ready, Completed).
- **Premium Membership:** Users can subscribe to premium plans for special discounts, faster service, and free on-campus delivery.
- **Notifications:** Real-time notifications for order confirmation, preparation, completion, and premium offers.
- **Theme Switch Mode:** Option to switch between Day and Night mode based on preference.
- **Payment Options:** Supports multiple payment methods – Cash on Pickup, UPI (Google Pay, Paytm, PhonePe), Digital Wallets, and Debit/Credit Cards.
- **Logout:** Ensures account security and safe session termination.

4.1.2 System Feature 2 – Admin Side (Functional Requirements)

The Admin Module assists the canteen management in order handling, record-keeping, and report generation, ensuring smooth daily operations.

Functional Requirements:

- **Dashboard:** Displays summary of users, active orders, food items, and recent activities with graphical sales reports.
- **Manage Food:** Admins can add, update, or delete food items, including categories, prices, and availability.
- **Manage Orders:** Allows viewing and updating of order status (Pending, Preparing, Ready, Delivered) and downloading receipts.
- **Manage Users:** Admins can view, edit, or deactivate user accounts, view feedback, and resolve user queries.
- **Help Desk:** View and respond to user-submitted support tickets. Resolved issues are archived.
- **Reports:** Generate daily, weekly, and monthly reports on sales, payments, and order summaries. Reports downloadable in PDF format.
- **Logout:** Allows the administrator to log out securely.

4.2 Hardware Requirements

Component	**Minimum Requirement**	**Recommended Requirement**
Processor	Intel Core i3 / AMD Equivalent	Intel Core i5 or Higher
RAM	4 GB	8 GB or Higher
Hard Disk	500 GB	1 TB
Display	15-inch Monitor	Full HD Monitor
Network	Stable Internet Connection	5 Mbps or Above
Input Devices	Keyboard, Mouse	Keyboard, Mouse, Touchscreen (optional)

Table : 2

4.3 Software Requirements

Software	**Requirement**
Operating System	Windows / Linux / macOS
Frontend	HTML5, CSS3, JavaScript, Bootstrap
Backend	Flask (Python)
Database	MySQL
Code Editor	Visual Studio Code / PyCharm
Web Server	XAMPP / Flask Built-in Server
Browser	Google Chrome / Mozilla Firefox
Plugins & Libraries	Flask, Bootstrap, jQuery, Chart.js

Table : 3

Justification:

- Flask ensures flexibility in backend integration with MySQL.
- Bootstrap enables responsive design across devices.
- Chart.js provides visual and interactive data reports.
- MySQL ensures structured, secure data storage and access.

4.4 Other Requirements

- The website must be mobile-responsive and compatible with all major browsers.
- User passwords should be encrypted for secure authentication.
- The system should support a minimum of 50 concurrent users.
- Response time should be less than 3 seconds per request.
- Reports and receipts should be downloadable in PDF format.
- Dark/Light mode should remember user preferences.

4.5 Database Requirements

Database Used: MySQL
MySQL manages users, menu items, orders, payments, favorites, notifications, and helpdesk records.

Major Tables:

- **users** – Stores user credentials and profiles.
- **admins** – Stores admin login credentials and roles.
- **categories** – Lists menu categories.
- **food_items** – Stores food details, prices, and availability.
- **cart** – Temporarily stores selected user items.
- **orders** – Contains order information and current state.
- **order_items** – Links each order to the selected food items.
- **payments** – Maintains payment details and modes.
- **favorites** – Stores user's favorite food items.
- **notifications** – Handles real-time updates for users.
- **helpdesk_tickets** – Logs issues and user complaints.
- **subscriptions** – Tracks premium membership details.
- **ratings** – Stores user feedback and ratings.

Relationships:

- One user may place several orders.
- Each order can include multiple food items.
- Admin manages food items, users, and reports.
- Helpdesk tickets are linked to users and resolved by admins.

4.6 Analysis Model

4.6.1 Data Flow Diagrams (DFD)

DFD LEVEL 0

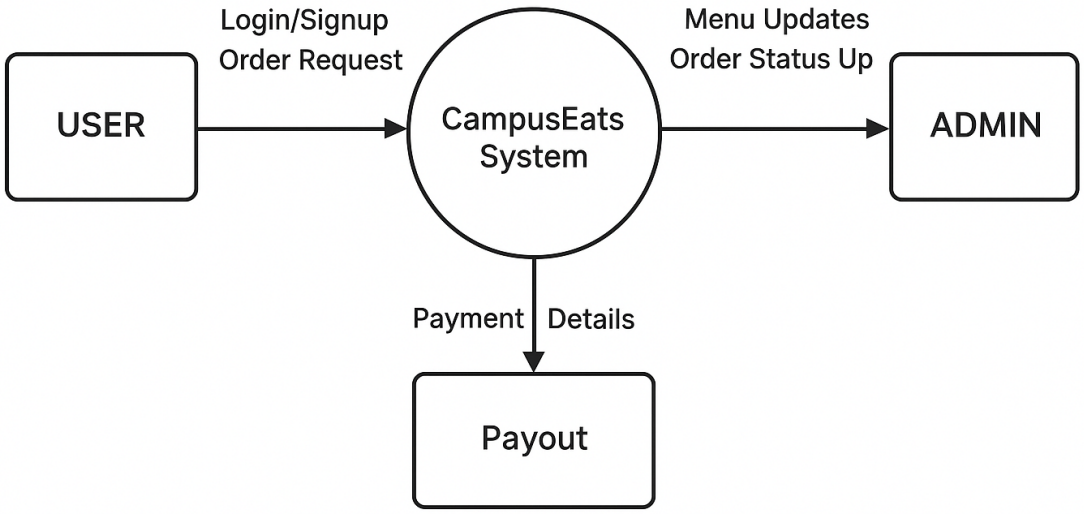


Fig: 1

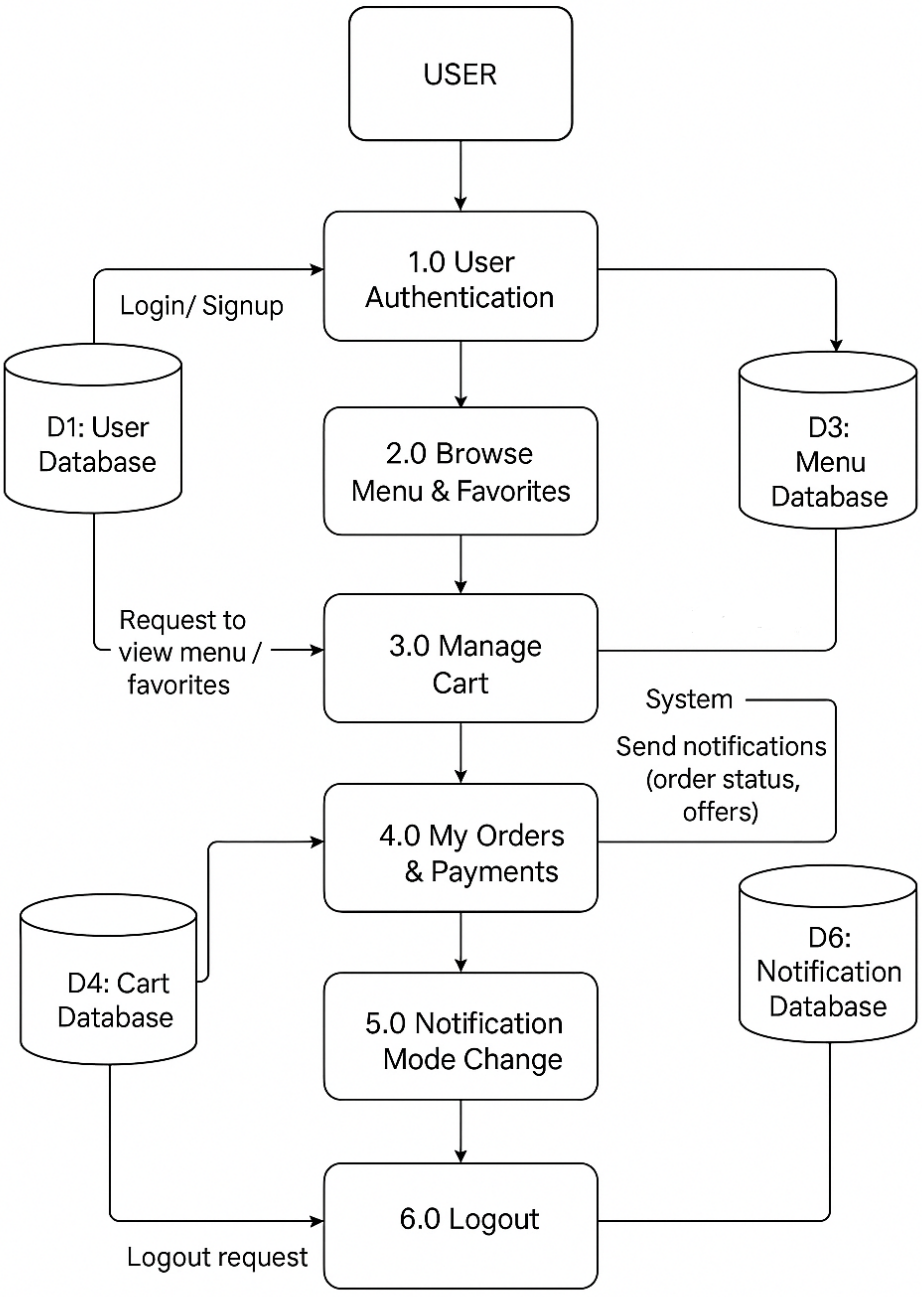

DFD Level 1 – User Side
USER
1.0 User Authentication
Login/ Signup
D1: User Database
2.0 Browse Menu & Favorites
D3: Menu Database
Request to view menu / favorites
3.0 Manage Cart
System
Send notifications (order status, offers)
4.0 My Orders & Payments
D4: Cart Database
D6: Notification Database
5.0 Notification Mode Change
6.0 Logout
Logout request

Fig: 2

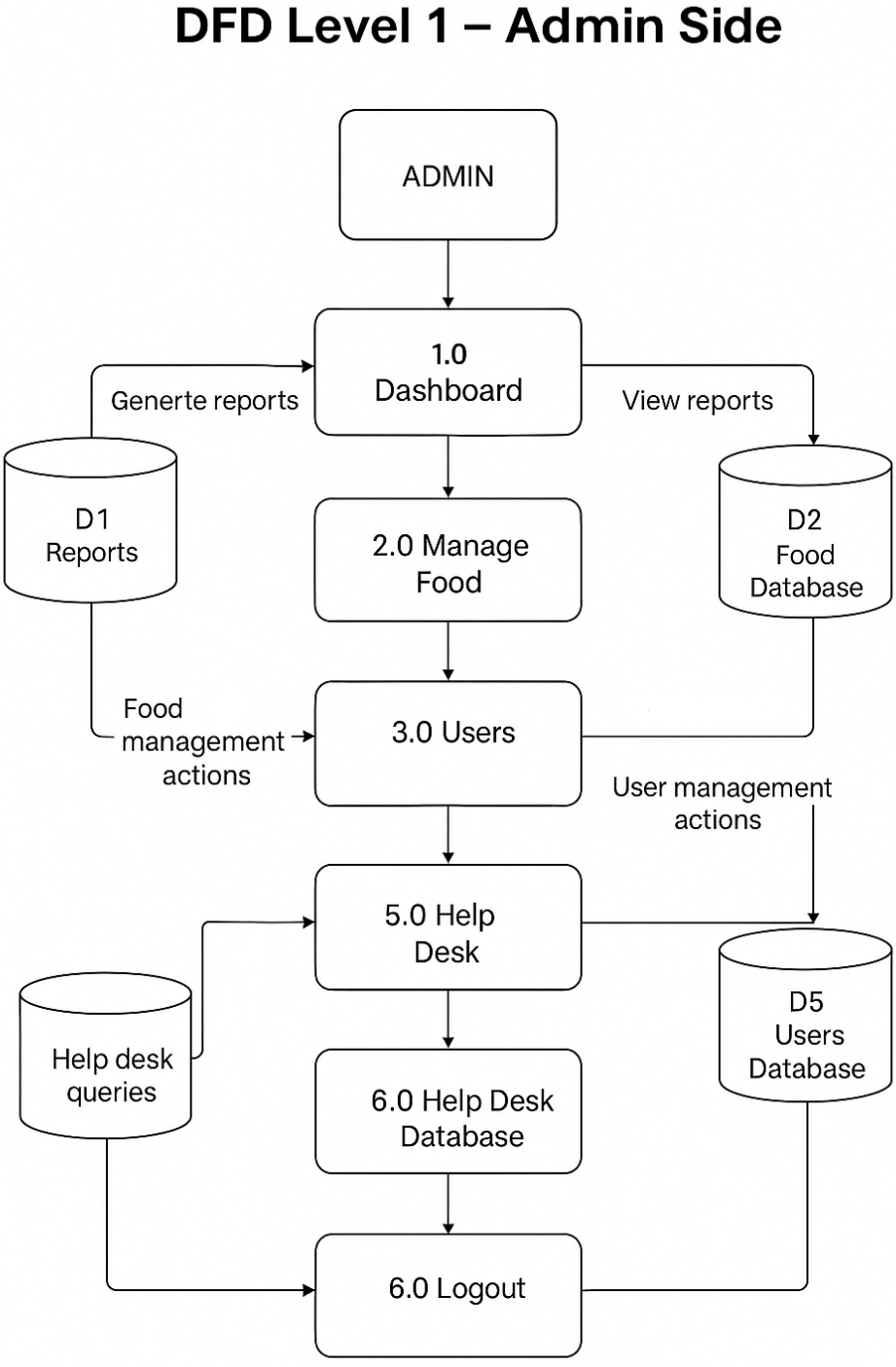

DFD Level 1 – Admin Side
ADMIN
1.0
Dashboard
Generte reports
View reports
D1
Reports
2.0 Manage
Food
D2
Food
Database
Food
management
actions
3.0 Users
User management
actions
5.0 Help
Desk
Help desk
queries
D5
Users
Database
6.0 Help Desk
Database
6.0 Logout

Fig: 3

4.6.2 Entity Relationship Diagram (ERD)

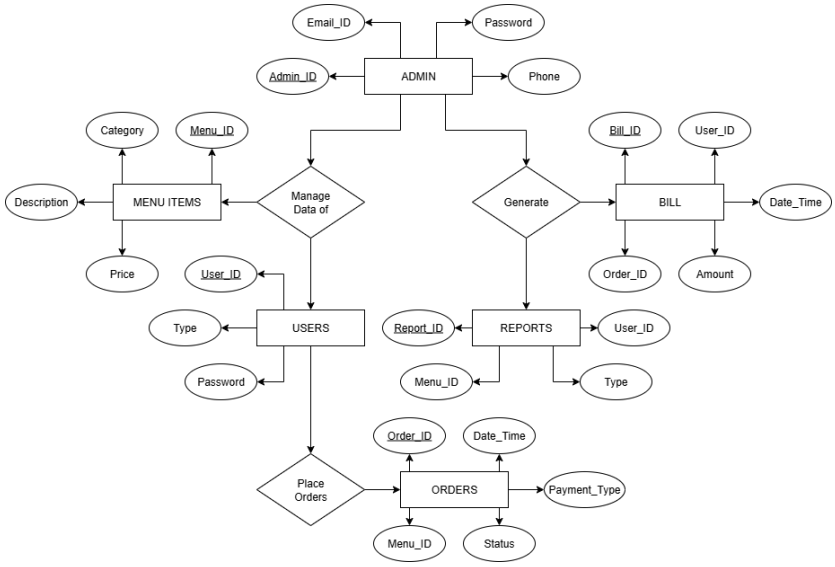


Fig: 4

4.6.3 UML Diagrams

4.6.3.1 Use Case Diagram

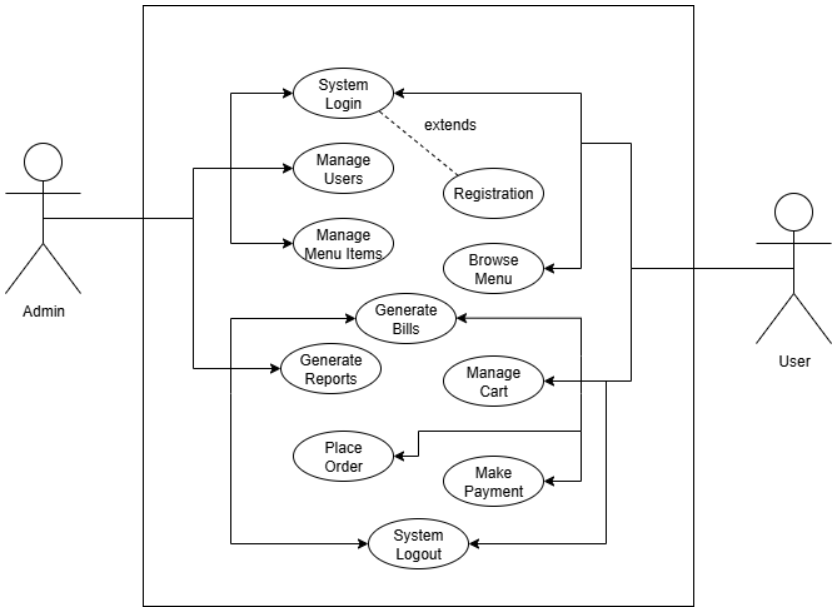


Fig: 5

4.6.3.2 Sequence Diagram

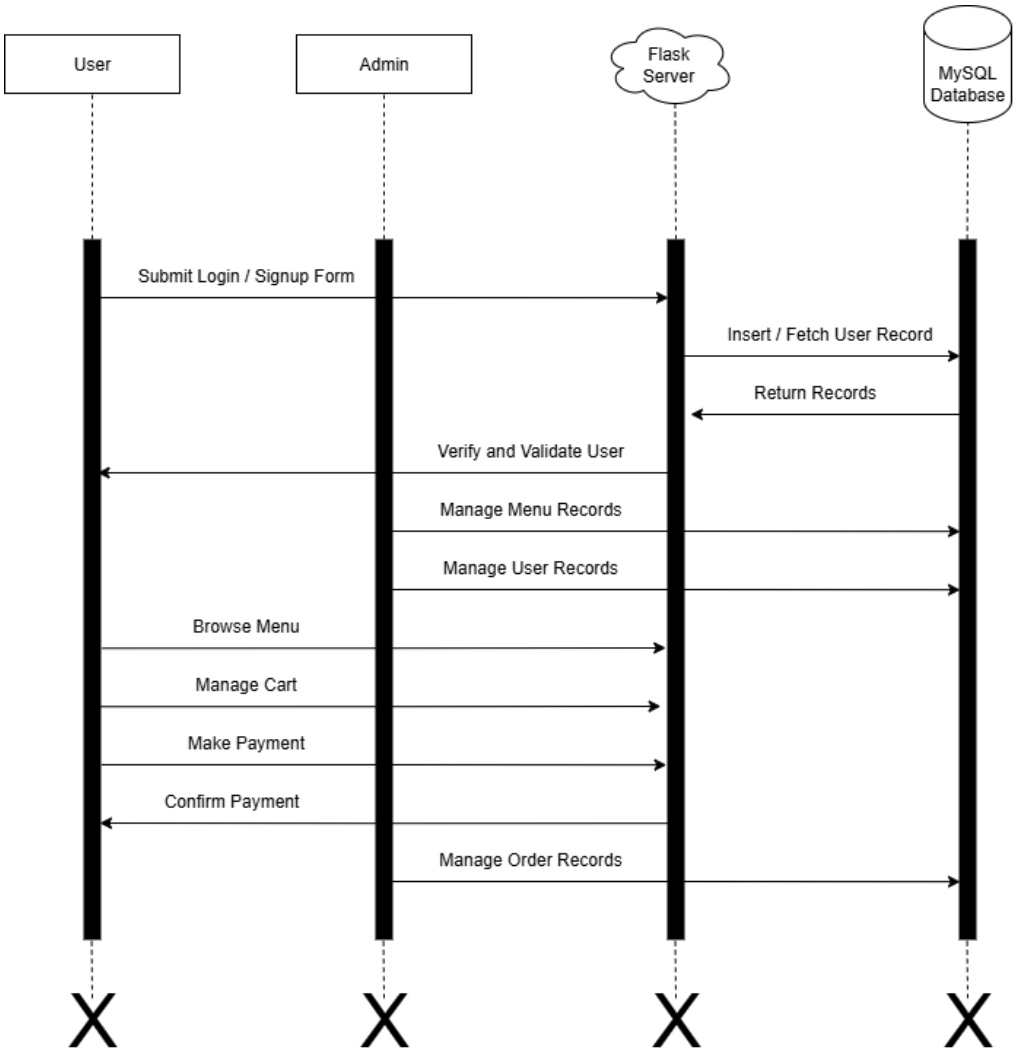


Fig: 6

4.6.3.3 Activity Diagram

Admin Activity Diagram

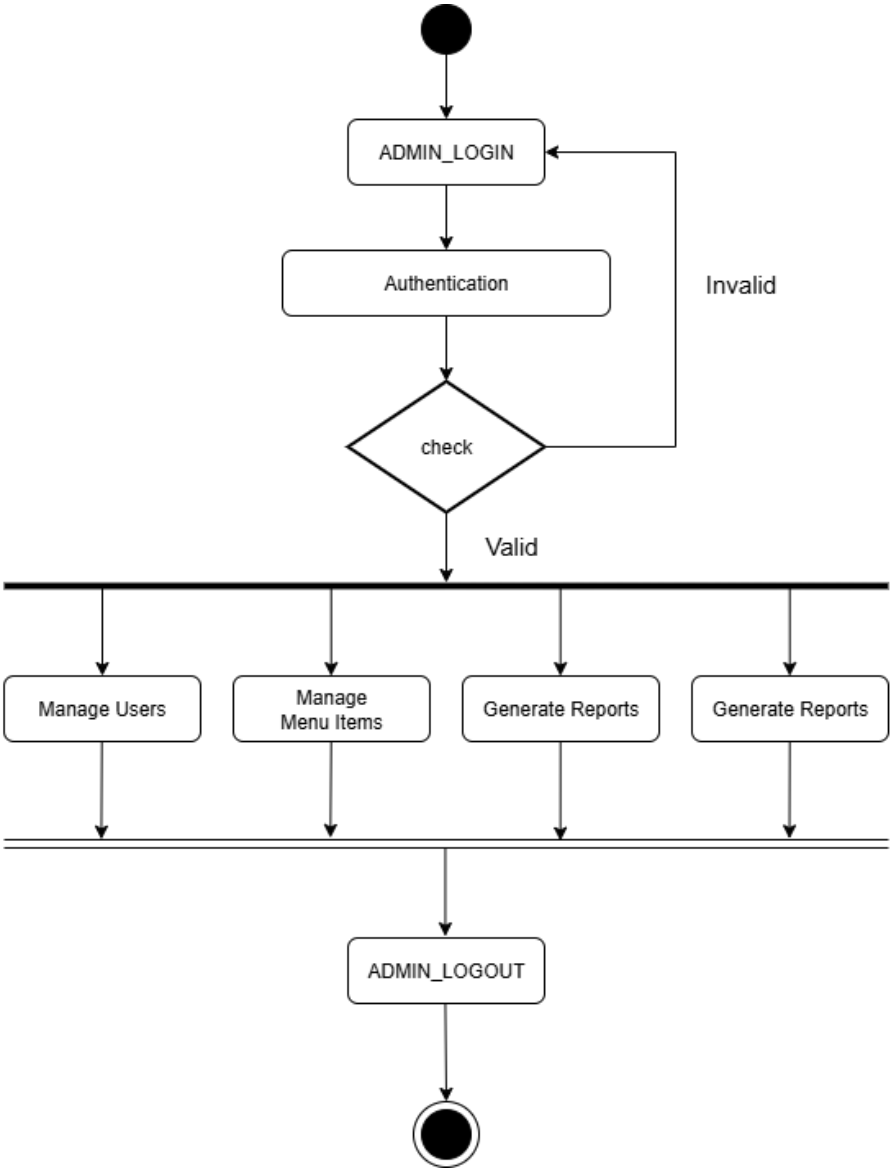


Fig: 7

User Activity Diagram

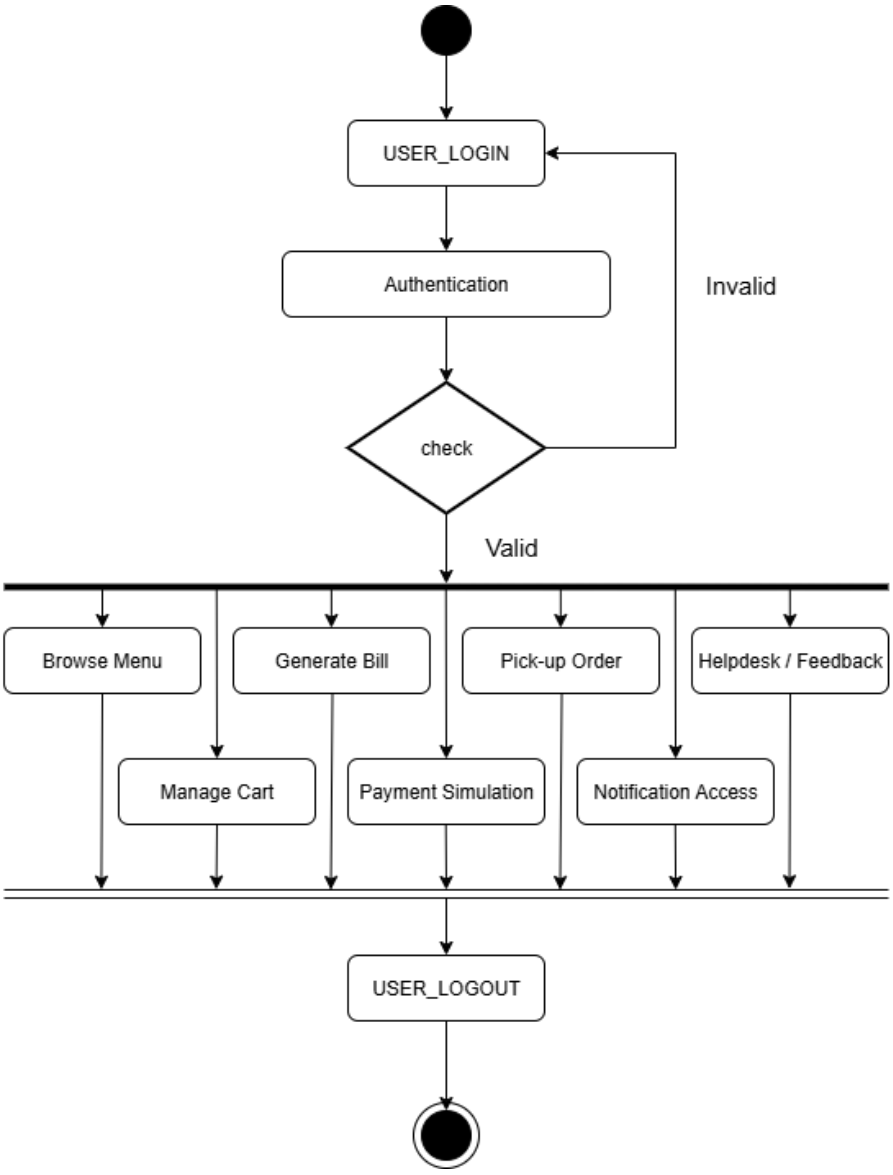


Fig: 8

4.6.3.4 Class Diagram

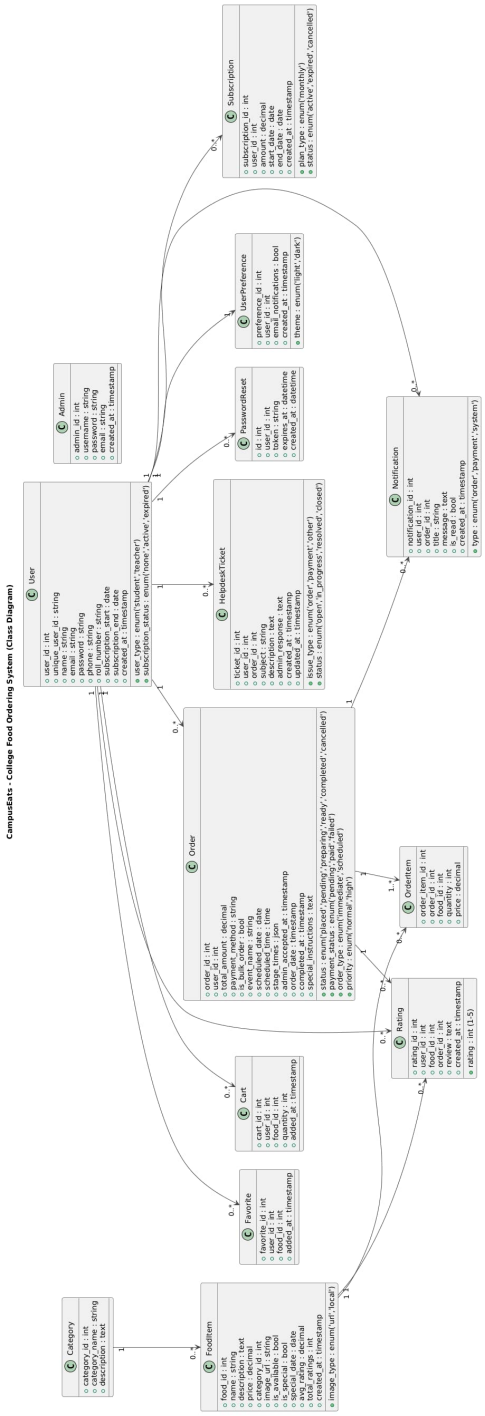


Fig: 9 Class Diagram

CHAPTER 5

System Analysis

5.1 System Architecture

The **CampusEats – Online Canteen Food Ordering For MET Campus** follows a **three-tier architecture** that separates the presentation, application, and database layers. This architecture ensures modularity, scalability, and maintainability, allowing independent updates to each layer without affecting the others.

5.1.1 Presentation Layer (Frontend / User Interface Layer)

This is the visual interface through which both users and administrators interact. It is developed using **HTML5, CSS3, JavaScript, and Bootstrap** for a responsive and user-friendly experience.

Key Features:

- Browse food items and categories.
- Add to cart, mark favorites, and track orders.
- Switch between light and dark modes.
- Make secure payments via multiple options.

Admin Interface Includes:

- Dashboard overview of users, orders, and items.
- Manage food items, menu, and order records.
- Generate reports and track sales data.

5.1.2 Application Layer (Logic / Middleware Layer)

This layer handles all the core business logic of the system and acts as a bridge between the frontend and the database. It is built using the **Flask framework in Python**.

Functions include:

- User authentication and authorization.
- Order management and payment validation.
- Dynamic menu fetching and updates.
- Real-time notifications and order tracking.

5.1.3 Database Layer (Data Management Layer)

This layer manages data storage and retrieval using **MySQL**. It stores user details, menu items, orders, payments, and reports.

Key Functions:

- Maintain integrity using primary and foreign keys.
- Support CRUD operations efficiently.
- Provide fast and accurate SQL queries.

5.1.4 Advantages of the Architecture

- **Modular Design:** Easier maintenance and scalability.
- **Security:** Isolated layers enhance safety.
- **Flexibility:** Easy integration of future modules.
- **Reliability:** Reduces redundancy and errors.

5.2 Modules

The system is divided into two main modules — **User Module** and **Admin Module**. Each module ensures smooth coordination between students, teachers, and canteen staff.

5.2.1 User Module

This module allows students and teachers to interact with the canteen system easily and efficiently.

Functions include:

- **Registration and Login:** Secure user sign-up and sign-in.
- **Menu Browsing:** View food categories and items dynamically.
- **Favorites:** Save favorite dishes for quick access.
- **Cart and Order Placement:** Add, edit, or delete items before confirming.
- **Order History:** View previous and current orders.
- **Notifications:** Receive real-time updates on order status.
- **Payment:** Pay using UPI, Wallet, or Cash on Pickup.
- **Premium Option:** Get discounts or priority services.

- **Logout:** Securely end user sessions.

5.2.2 Admin Module

The Admin Module enables canteen staff to manage and monitor operations digitally.
Functions include:

- **Admin Login:** Secure authentication for authorized users.
- **Dashboard:** Overview of all user and order activities.
- **Manage Menu:** Add, update, or delete food items.
- **Manage Orders:** Track and update order statuses.
- **User Management:** View and manage user accounts.
- **Help Desk:** Handle complaints and feedback.
- **Report Generation:** Export order and sales reports as PDF.
- **Logout:** Securely close admin session.

5.3 Algorithm / Techniques Used

To ensure smooth performance and secure processing, the system implements several efficient algorithms and techniques.

5.3.1 Priority Queue Algorithm for Order Processing

Orders are managed using a **priority queue** system, giving teachers higher priority.
Process:

1. Each order receives a priority:
    - Teacher → Priority 1
    - Student → Priority 2
2. Orders are queued and processed based on priority.

5.3.2 Real-Time Order Tracking Mechanism

The tracking feature updates the order status dynamically.
Process:

1. Orders start with status *Pending*.
2. Status changes to *Preparing*, *Ready*, and finally *Completed*.

3. Each status update triggers a user notification.

5.3.3 Payment Validation and Security

The payment system validates every transaction securely.
Steps:

1. User selects payment method.
2. System generates a unique transaction ID.
3. Payment details are encrypted.
4. On success, the order is marked as *Completed*.

5.3.4 Data Encryption and Security Techniques

- Passwords are hashed using SHA or MD5.
- Parameterized SQL queries prevent injection attacks.
- Sessions are securely managed.
- Error handling ensures data reliability.

5.3.5 Report and Receipt Generation Technique

Admins can generate and download dynamic PDF reports of all orders. Flask fetches real-time order data and uses templates for report creation. Each order is assigned a unique **Receipt ID** for easy tracking.

CHAPTER 6
Project Plan

6.1 Stakeholder List

A stakeholder is any individual, group, or organization involved in or affected by the **CampusEats – Online Canteen Food Ordering For MET Campus.** Each stakeholder plays a critical role in ensuring smooth system operation, testing, and successful project completion.

Stakeholder	**Role / Responsibility**	**Interest in System**
Project Developers (Team Members)	Design, code, and test the project using Flask, MySQL, HTML, and CSS. Ensure the system is user-friendly and secure.	High – Responsible for system creation and maintenance.
Project Guide (Prof. Mehraj Khan)	Supervises technical development, reviews progress, and ensures correct project methodology.	High – Provides direction and technical mentorship.
Head of Department (Prof. P. D. Jadhav)	Evaluates project deliverables and approves for academic submission.	Moderate – Ensures departmental quality compliance.
Canteen Administrator (Admin User)	Manages menu items, handles orders, user data, and reports.	Very High – Uses the system daily for operations.
Students and Teachers (End Users)	Use the CampusEats website to browse menus, place orders, and make payments.	Very High – Primary beneficiaries of the system.
Database Administrator (Developer Team)	Maintains and secures the MySQL database, manages backups, and prevents data loss.	High – Ensures smooth backend operations.
Institution (MET BKC IOE)	Provides infrastructure support, project evaluation, and deployment approval.	Moderate – Facilitates campus-level implementation.

Table: 4

6.1.1 Stakeholder Relationships

- **Developers Guide:** Guidance, regular reviews, and technical corrections.
- **Developers Admin:** Coordination for menu and order management testing.
- **Admin Users:** Handles food orders and service interactions.
- **Institution Team:** Provides infrastructure and hosting support.

6.2 System Implementation Plan

6.2.1 Main Pages

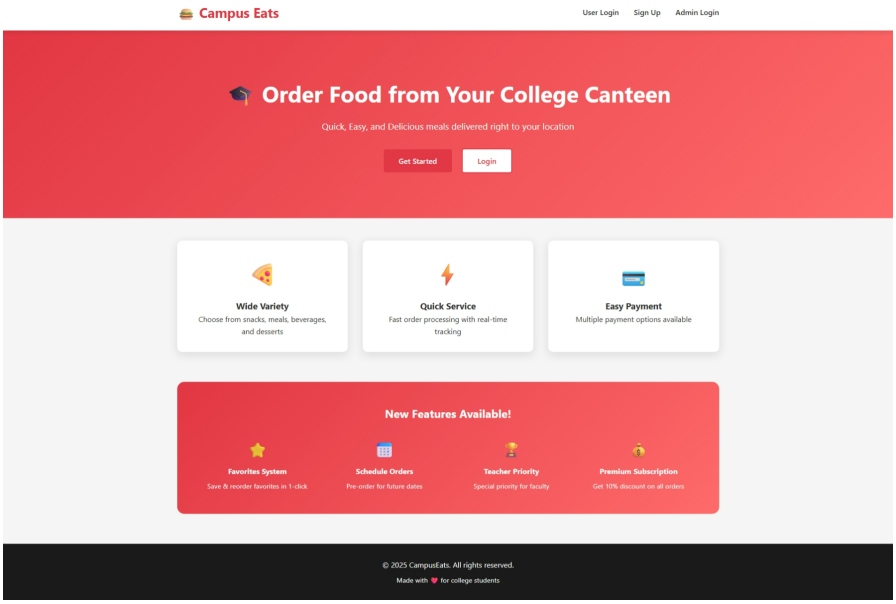


Fig : 1 - Home Page

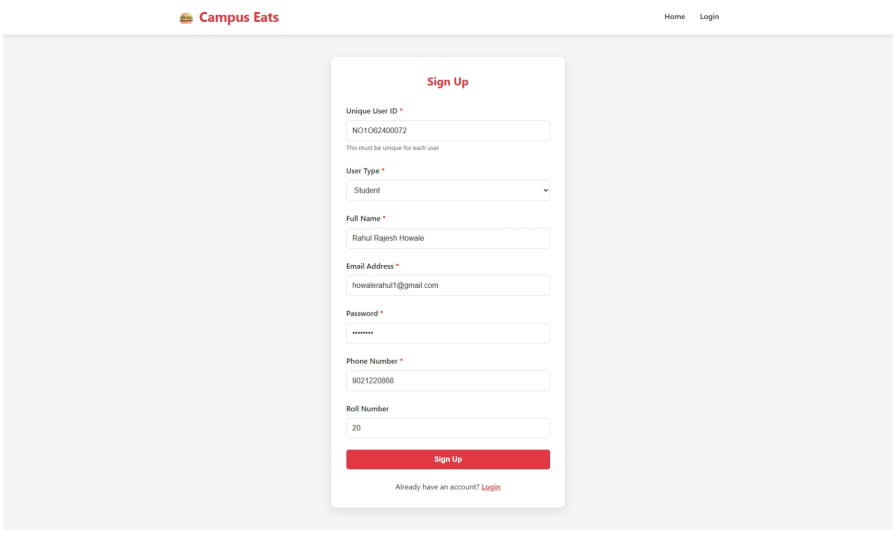


Fig : 2 - SignUp Page

6.2.2 Admin Side

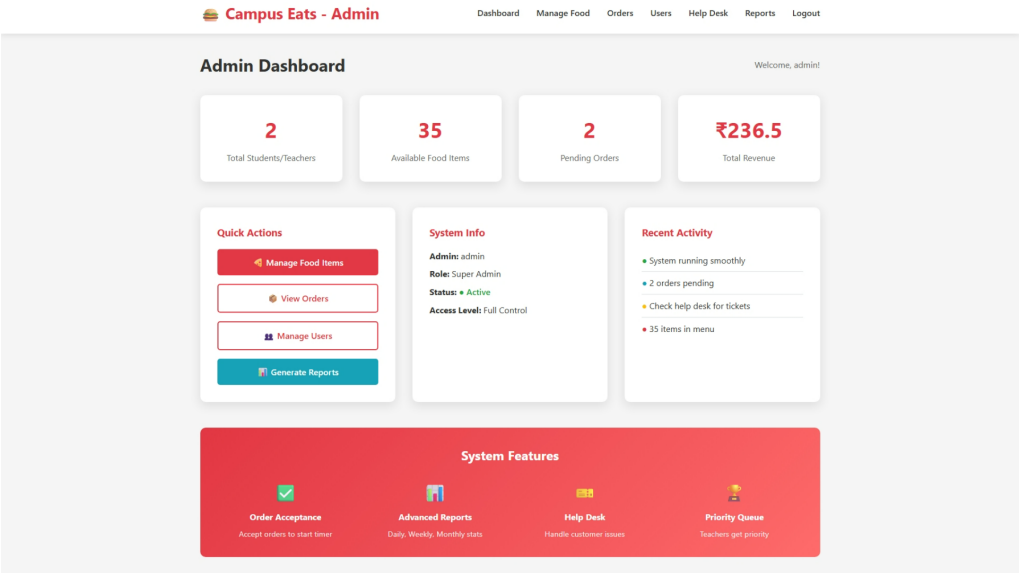


Fig : 3 - Admin Dashboard

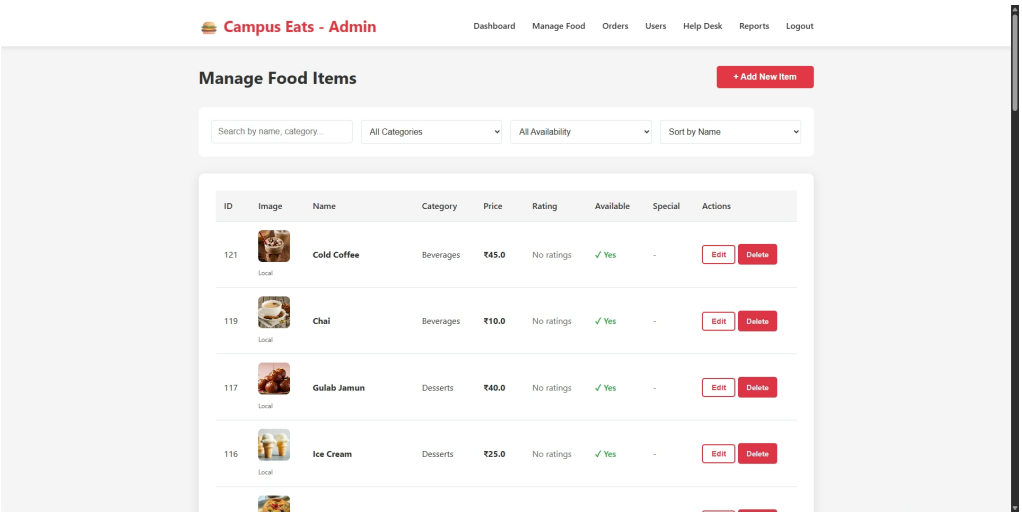


Fig : 4- Manage Food Orders

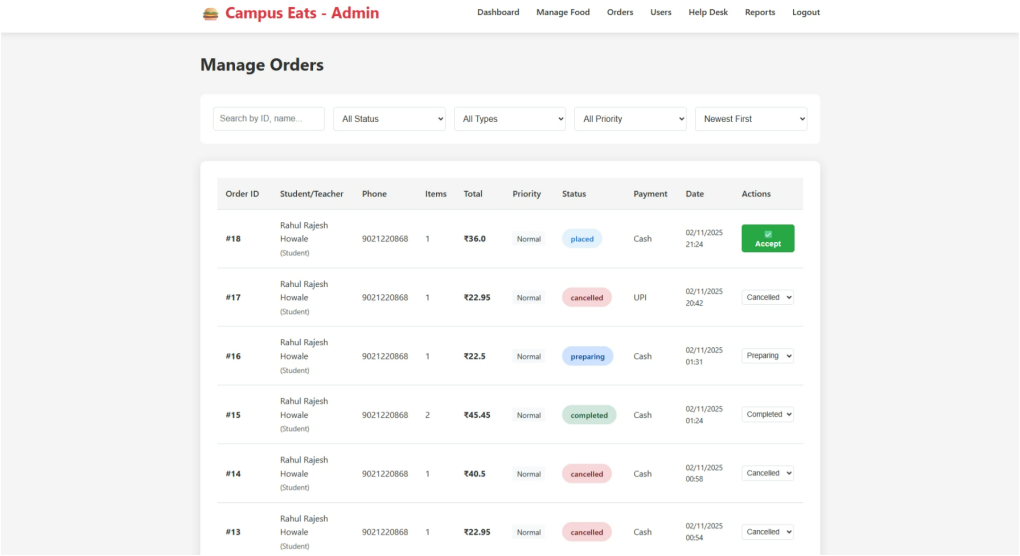


Fig : 5 - Manage Orders

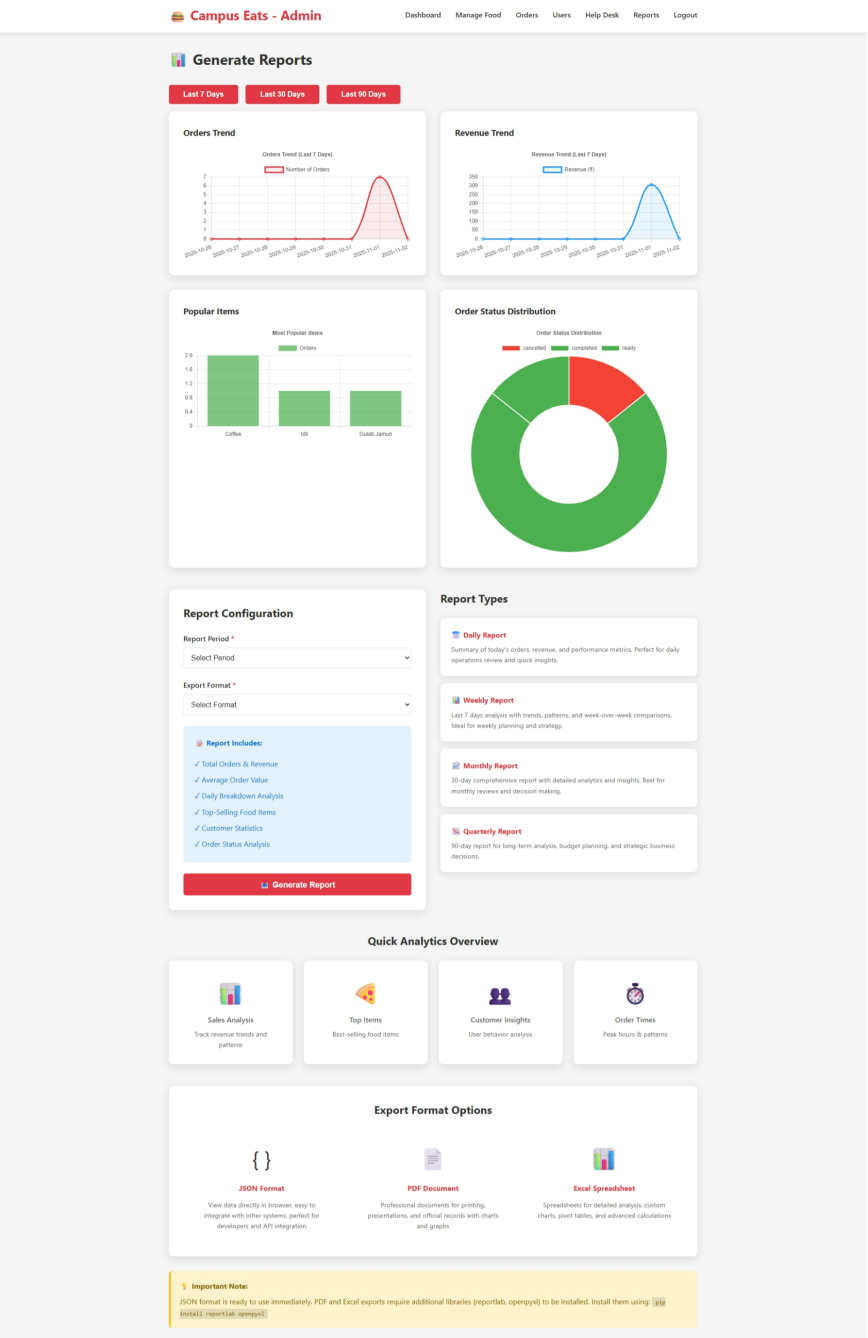


Fig : 6 - Reports

6.2.3 User Side

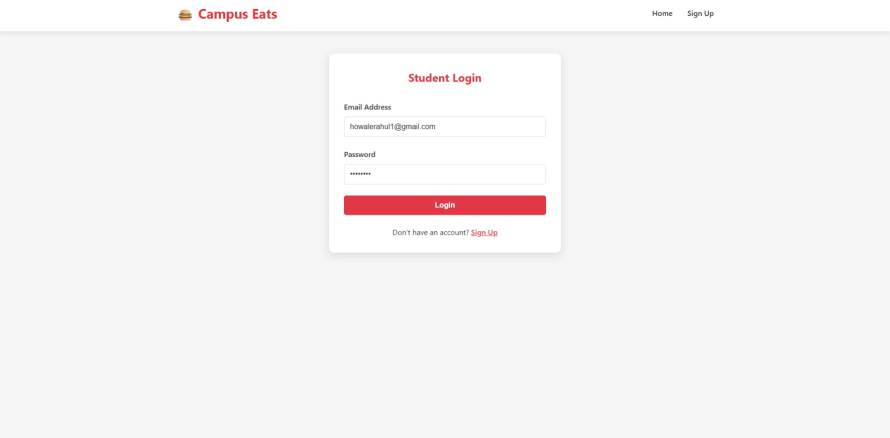


Fig : 7 - User Login Page

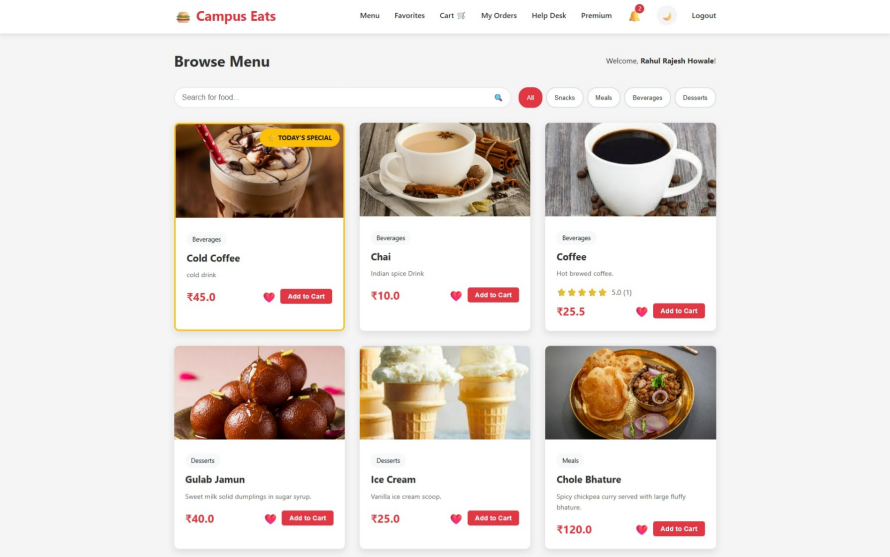


Fig : 8 - Browse Menu

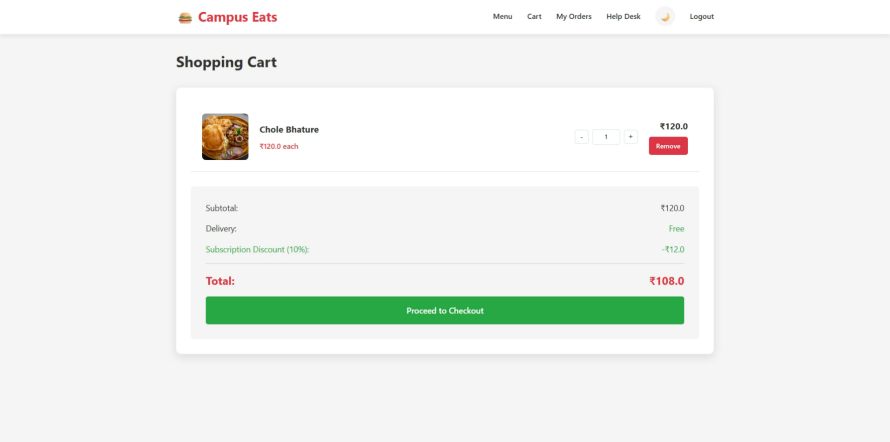


Fig : 9 - Shopping Cart

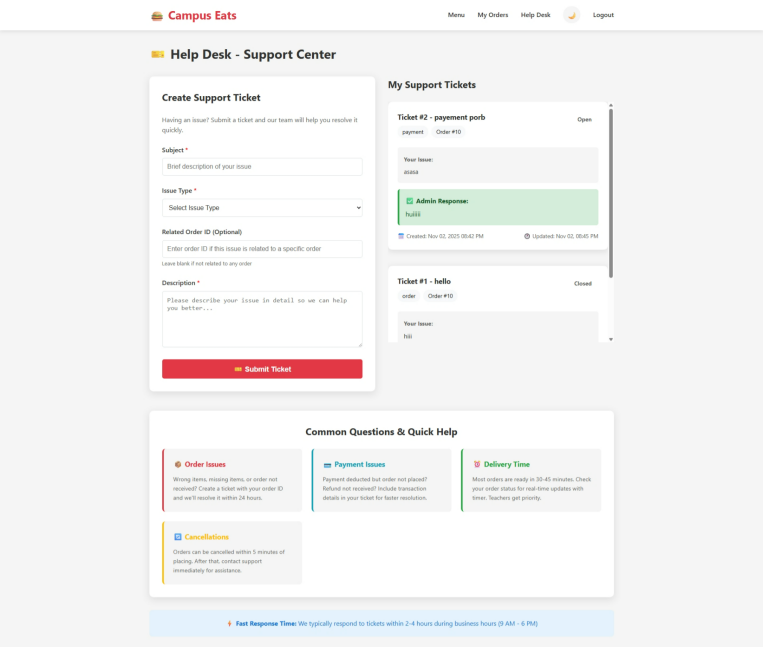


Fig : 10 - Help Desk

CHAPTER 7

Conclusion

The **CampusEats - Online Canteen Food Ordering For MET Campus** eliminates the inefficiencies and limitations found in traditional canteen management systems and introduces a modern, digital approach with convenience at its core. This project integrates all major functionalities — from browsing the menu to placing orders, making payments, and managing food items — into a single, unified web application.

It follows a **three-tier architecture**, divided into the presentation, application, and database layers, ensuring modularity, scalability, and maintainability. The backend is developed using **Flask (Python)**, while the frontend employs **HTML, CSS, JavaScript, and Bootstrap** for an interactive user experience. **MySQL** efficiently handles the database operations, maintaining data integrity, reliability, and security.

From the **user's perspective**, the CampusEats platform offers a smooth and intuitive interface for students and teachers. They can browse the menu, mark favorite items, track real-time orders, and make secure online payments using multiple modes such as UPI, wallets, or cards. Features like **dark/light theme switching** and **premium subscriptions** enhance personalization and user engagement.

From the **administrator's viewpoint**, the system simplifies canteen operations through an intuitive dashboard that manages users, orders, and menu items. The admin can generate **sales reports**, view order analytics, and address help desk queries easily. Additionally, the use of a **priority queue mechanism** ensures that teachers receive faster service, promoting fairness and efficiency during rush hours.

A strong emphasis has been placed on **security, authentication, and transparency**. Password encryption, secure session management, and proper input validation prevent unauthorized access and safeguard user data, maintaining trust between the system and its stakeholders.

In conclusion, the **CampusEats project** demonstrates how technology can transform traditional campus operations into efficient, digital processes. It minimizes manual workload, reduces waiting time, and strengthens communication among students, teachers, and canteen staff. More than just an ordering platform, it embodies the vision of a **"Smart Campus Ecosystem."**

The system is highly scalable, supporting potential future enhancements such as:

- Integration with Android and iOS mobile applications.
- AI-powered food recommendations based on user preferences.
- QR-based pickup systems for contactless delivery.
- Integration with college ID or wallet systems for easy payments.

Overall, **CampusEats** stands as a testament to teamwork, innovation, and modern web development practices. It not only streamlines canteen operations but also sets a new standard for campus digitization, contributing meaningfully to the broader vision of smart institutional infrastructure.

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